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Renal excretion of meta-iodobenzylguanidine after therapeutic doses in cancer patients and its relation to dose and creatinine clearance

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Summary

Iodine-131 radioiodinated meta-iodobenzylguanidine (^{131}I-MIBG) is used for diagnostic scintigraphy and radionuclide therapy of neural crest-derived tumours. In higher doses (up to 80 mg), non-radioactive MIBG is now evaluated for palliation in carcinoid patients. After administration of therapeutic doses of ^{131}I-MIBG (3.7–7.4 GBq, 1.7–5.8 mg MIBG) to patients aged 2–73 years, 53 ± 8.8%, 69 ± 7.8% and 83 ± 7.0% of the dose was cumulatively excreted as MIBG in the urine after 24, 48 and 72 h, respectively. Within the MIBG dose range of 1.7–80.0 mg, a linear relationship was found between the excretion rate over the first 24 h (mg per 24 h) and the dose. In adults, the MIBG excretion rate over the first 24 h (% of dose per 24 h) was shown to be only partially related to the glomerular filtration rate (GFR).

Introduction

At present, the use of radioiodinated meta-iodobenzylguanidine (^{131}I-MIBG or ^{123}I-MIBG) occupies a well-established position in diagnostic scintigraphy of neural crest-derived tumours, such as neuroblastoma and malignant phaeochromocytoma [1]. Moreover, ^{131}I-MIBG is also employed in the radionuclide therapy of such tumours [1–4], whereas at the Netherlands Cancer Institute non-radioactive MIBG is currently administered to carcinoid patients in order to evaluate its possible contribution to the palliative effects of ^{131}I-MIBG [5]. Despite its wide use, relatively little has been published on the human pharmacokinetics of ^{131}I-MIBG [6]. However, it is known that, following intravenous administration, radioactivity is recovered almost entirely in urine [6]. Moreover, it has been shown that ^{131}I-MIBG is excreted largely unchanged in urine [7, 8]. Therefore, determination of the urinary excretion pattern of unchanged MIBG or ^{131}I-MIBG is a valid and non-invasive way of gaining further insight into the pharmacokinetics of MIBG. Whereas a therapeutic ^{131}I-MIBG dose of 3.7–7.4 GBq is equal to 3.4–6.7 mg MIBG, doses of up to 80 mg MIBG were given in a non-radioactive study [5].

We reported previously the development of a high-performance liquid chromatographic (HPLC) procedure with on-line UV detection for the analysis of MIBG in the urine samples of patients undergoing treatment [9]. This procedure enables us to study the urinary pharmacokinetics of MIBG after the administration of ^{131}I-MIBG or MIBG. The glomerular filtration rate (GFR) has been reported to exert a major influence on the pharmacokinetics of (diagnostic doses of) ^{123}I-MIBG in adults [10]. The present study aimed to investigate the urinary pharmacokinetics of therapeutic doses of MIBG, administered as either ^{131}I-MIBG or MIBG to adults and children suffering from different neural crest-derived tumours. Moreover, the relationships between MIBG dose and excretion rate and between MIBG excretion rate and creatinine clearance (as a measure of the GFR) were evaluated.

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Patients and methods

Patients

All patients were selected for either MIBG or ^{131}I-MIBG therapy after a scintigraphic assessment of tumour uptake and retention using 37 MBq of ^{131}I-MIBG (specific activity >1.11 GBq mg^{-1}). The characteristics of the patients in this study are shown in Table 1. All patients gave their informed consent for the collection of urine for research purposes. Some of the patients (J.C., V.B., C.O.) were studied more than once.

Materials

^{131}I-MIBG·½H_2SO_4 [glass vials with 3.7 or 1.85 (±10%) GBq in 7.5 ml 0.9% (w/v) sodium chloride solution containing 1% (w/v) benzyl alcohol; specific activity at reference = 1.2–2.1 GBq mg^{-1} (0.39–0.68 TBq mmol^{-1})] was purchased from Amersham Buchler (Braunschweig, Germany). The specific activity, radionuclidic and radiochemical purity of each ^{131}I-MIBG batch were assessed and met the specifications of the manufacturer (i.e. >1.11 GBq mg^{-1}, ≥99.9% and ≥95%, respectively). The batches used in the present study were shown to contain 95.2–97.6% ^{131}I-MIBG and 0.1–0.3% ^{131}I-labelled meta-iodobenzylamine (^{131}I-MIBAM) using a HPLC procedure [11]. Pre-administration quality control showed the batches used contained 3.0–4.9% free ^{131}I-iodide, determined using a validated solid-phase extraction (SPE) procedure [11, 12].

MIBG·½H_2SO_4 was synthesized from the hydrochloride (HCl) salt of MIBAM by a modified version of the method of Wieland *et al.* [13]. Identification was by means of chemical ionization mass spectrometry (CI-MS), proton nuclear magnetic resonance spectroscopy (NMR) and infrared spectrophotometry (IR), yielding the following characteristics:

- MS (CI) m/z: 276 ($[M+H]^+$), 217 ($[IPhCH_2]^+$), 150 ($[PhCH_2NHCNHNH_2+H]^+$), 90 ($[PhCH_2]^+$), 60 ($[NH_2CNHNH_2+H]^+$).
- NMR: delta 9.10 (1H, m, CH_2NHC), 7.80 (3H, s, $CNHNH_2$), 7.70 (1H, s, CH-arom.), 7.65 (1H, m, CH-arom.), 7.40 (1H, m, CH-arom.), 7.20 (1H, m, CH-arom.), 4.35 (2H, m, CH_2).
- IR (KBr) cm^{-1}: 3320 and 3140 (NH), 1660 and 1640 (C=N), 1100 (SO_4), 770 and 690 (1,3-disubstituted benzene).

The chemical purity of MIBG was determined by means of the HPLC procedure as described earlier [9] and was

Table 1. Characteristics of the patients.

Patient	*Sex*	*Age (years)*	*Weight (kg)*	*Body surface (m^2)*	*Tumour type*	*Co-medication*[a]
^{131}I-MIBG therapy, children						
D.J.	M	5	N.A.	N.A.	Neuroblastoma	None
N.Z.	M	3	14	N.A.	Neuroblastoma	None
Z.L.	M	2	13	0.58	Neuroblastoma	None
T.R.	M	8	34	1.14	Neuroblastoma	None
L.N.	F	5	17	0.72	Neuroblastoma	None
I.S.	F	11	36	1.21	Neuroblastoma	None
^{131}I-MIBG therapy, adults						
M.G.	F	73	56	1.60	Carcinoid	Cholestyramine, loperamide, morphine
J.C.	M	51	76	1.88	Phaeochromocytoma	Phenoxybenzamine, propranolol, ranitidine
A.G.	M	64	79	2.04	Carcinoid	Acetylcysteine, ipratropium, levothyroxine, methysergide, salbutamol
V.B.	M	39	57	1.68	Phaeochromocytoma	Phenoxybenzamine, propranolol
R.R.	F	40	74	1.79	Neuroblastoma	None
MIBG therapy, adults						
C.O.	F	65	46	1.42	Carcinoid	Digoxin, octreotide
C.V.	F	62	61	1.67	Carcinoid	Ketanserin, temazepam
E.K.	F	64	N.A.	N.A.	Carcinoid	N.A.
J.B.	M	44	77	1.98	Carcinoid	Loperamide, potassium chloride
B.A.	M	50	62	N.A.	Carcinoid	Ketanserin, methadone, morphine

[a]All patients on ^{131}I-MIBG therapy ingested 200 mg potassium iodide once daily (children under 14 years: 100 mg). In addition, the patients on ^{131}I-MIBG occasionally used domperiodone or metoclopramide; the patients on MIBG therapy occasionally used acetaminophen. N.A., not available.

found to be >99% with <1% MIBAM. MIBAM·HCl was purchased from Janssen Chimica (Geel, Belgium).

Drug administration and collection of urine samples

Therapeutic doses of ^{131}I-MIBG (3.7–7.4 GBq, 1.7–5.8 mg MIBG) in 100 ml 0.9% (w/v) sodium chloride solution [11], or MIBG (14–80 mg) in 100 ml 0.9% (w/v) sodium chloride solution, were administered as a 4-h intravenous infusion. A urine sample voided prior to the start of infusion was taken as the blank. The urine of patients treated with MIBG was collected during the first 24 h after the start of infusion. The total urinary output of patients treated with ^{131}I-MIBG was collected until discharge from hospital (4–5 days after infusion). Urine collection after administration of ^{131}I-MIBG was performed at the end of infusion, twice daily for the following 2 days and, subsequently, at 24-h intervals. For each sample, the amount of urine excreted was read from calibrated containers and a 20-ml sample was taken for analysis. The samples were stored at −25°C (samples provided at the weekend: ≤3 days at 4°C and subsequently at −25°C) from the moment of collection until analysis (1 month).

Analysis

Determination of MIBG in urine was performed as described elsewhere [9]. The completeness of each urine sample of the ^{131}I-MIBG treated patients was checked by interview and by comparing the amount of radioactivity collected (and corrected for radioactive decay) with the corresponding values calculated from exposure rate ($\dot{X}$) measurements. When, after infusion and before the first voiding, the activity in a patient (A, expressed in MBq) is known and $\dot{X}$ (in μSv h^{-1}) is measured at l = 1 m distance from the patient, the specific gamma-ray constant (Γ, in μSv m^2 MBq^{-1} h^{-1}) for that patient follows from $\dot{X} = A * \Gamma * l^{-2}$. Subsequently, Γ can be used to calculate the remaining activity in, and thus the excreted activity from, the patient at later time points by measuring $\dot{X}$ at 1 m distance from the patient. Cumulative radioactivity in the urine of patients who confirmed complete urine collections during hospitalization ($U_{cum,total}$) amounted to 95.8 ± 8.0% (mean ± S.D., n = 19) of the corresponding values ($E_{cum,total}$) calculated from exposure rate measurements. Based on this, for all urine samples of patients (n = 6, Table 2) with known incomplete collections (e.g. because of the use of diapers), the amounts of MIBG as determined by HPLC were multiplied by the corresponding E_{cum}/U_{cum} factor, which was calculated individually for each sample.

The elimination half-lives ($t_{½,elim}$) of MIBG were calculated from semi-logarithmic plots of the average excretion rate of MIBG (mg h^{-1}) per urine collection versus the midpoint time of urine collection, a procedure described by Rowland and Tozer [14].

Creatinine clearance values (Cl_{cr}) were calculated from a recent (within 1 week from MIBG or ^{131}I-MIBG administration) serum creatinine determination. For adult patients, the equation of Cockroft and Gault [15] was used and multiplied by 0.85 for females so that they could be compared with males. Cl_{cr} values for children were calculated using the equation of Schwartz *et al.* [16].

Statistical analysis of the relationships between measured variables was performed using linear regression analysis in combination with Student's t-test for significance. Statistical significance was accepted at the 5% level.

Results

Table 2 and Fig. 1 show that cumulative excretion 24, 48 and 72 h after the start of infusion of a therapeutic dose of ^{131}I-MIBG containing 1.7–5.8 mg MIBG, amounted to 53 ± 8.8% (mean ± S.D., n = 14), 69 ± 7.8% (n = 10) and 83 ± 7.0% (n = 3), respectively. The MIBG elimination rate in the combined ^{131}I-MIBG adult and child patient groups over the first 44 h (four samples per patient) resulted in a $t_{½,elim}$ of 11.5 ± 1.7 h (n = 11). The children tended to eliminate MIBG slightly faster than the adults: $t_{½,elim}$ = 10.6±1.8 h for the children compared with 12.4 ± 1.3 h for the adults (Table 2), although the figures were not statistically significant.

The following highly correlated linear relationship was found between MIBG dose and excretion rate (expressed as 24-h cumulative excretion of MIBG) within the dose range 1.7–80.0 mg, for the patient population as a whole, including the children:

$$y = 0.367\ (\pm 0.026)x + 0.584\ (\pm 0.618)$$
$$(r = 0.9556,\ P<0.001,\ n = 21)$$

where y is the excretion rate of MIBG over the first 24 h (mg per 24 h) and x is the MIBG dose in mg (Table 2).

Furthermore, a statistically significant correlation (r = 0.7311, P<0.02) was found between the renal MIBG excretion rate, expressed as % dose per 24 h, and GFR in the adult patient population (n = 11). The children, who had higher 0–24 h MIBG excretion percentages and lower Cl_{cr} values compared with the adults (Table 2), did not fit the significantly correlated 0–24 h MIBG excretion percentage versus GFR plot of the adult patient population.

Discussion

The renal excretion pattern of MIBG after intravenous administration of a therapeutic dose of ^{131}I-MIBG (1.7–

Table 2. MIBG doses and excretion parameters.

Patient	Creatinine clearance (ml min^{-1})	MIBG dose (mg)	0–24 h renal MIBG excretion		0–48 h renal MIBG excretion (% of dose)	0–72 h renal MIBG excretion (% of dose)	$t_{½,elim}$[a] (h)
			mg	% of dose			
^{131}I-MIBG therapy, children							
D.J.	N.A.	1.7	0.94	55	78	86	13.6
N.Z.	N.A.	2.5	1.5	59	71	—	9.6
Z.L.[b]	65	3.8	2.2	57	69	—	9.9
T.R.[b]	80	3.9	2.8	72	83	88	9.1
L.N.[b]	42	4.4	2.5	57	69	—	[d]
I.S.	N.A.	4.9	2.5	52	65	—	10.6
^{131}I-MIBG therapy, adults							
M.G.[b,c]	61	1.8	0.72	40	—	—	—
J.C.	63	3.3	1.5	46	—	—	14.3
J.C.	N.A.	4.2	2.0	47	63	—	13.5
A.G.[b]	61	4.2	1.8	42	—	—	—
V.B.	77	4.6	2.7	58	73	—	11.4
J.C.	N.A.	4.7	2.5	54	67	75	11.8
R.R.[b]	N.A.	5.5	3.5	63	—	—	11.1
V.B.	86	5.8	2.6	44	55	—	12.0
MIBG therapy, adults							
C.O.	46	14.0	4.1	29			
C.V.	69	17.0	9.7	57			
E.K.	N.A.	18.0	6.3	35			
J.B.	74	19.0	9.5	50			
C.O.	50	29.0	10.0	35			
C.O.	47	57.0	14.0	25			
B.A.	83	80.0	35.0	44			

[a] $t_{½,elim}$: elimination half-life calculated from the first four urine collections covering 44 h from the start of infusion.
[b] Patients with one or more incomplete collections; the data were corrected using exposure rate measurements (see text).
[c] Administered intra-arterially in 3 h (arteria hepatica, catheterized via arteria femoralis).
[d] No calculation of $t_{½,elim}$ due to too few separate urine collections.
N.A., not available; —, no cumulative data over the indicated period, since MIBG concentrations in urine samples from later collections were below the lower limit of quantitation of the HPLC procedure (see text).

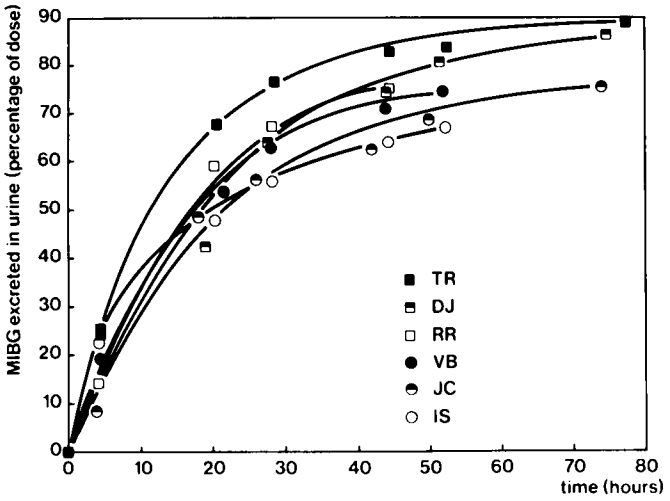


Fig. 1. Cumulative percentage of dose excreted in the urine versus time plots of different patients after the administration of therapeutic amounts of ^{131}I-MIBG (3.7–7.4 GBq, 1.7–5.5 mg MIBG).

5.8 mg MIBG) is relatively constant (Table 2, Fig. 1), which is more meaningful when one considers the diversity of the patient population studied (age range 2–73 years, different neural crest tumours, with or without other medications). In order to establish the relationship between MIBG dose and excretion rate within the dose range 1.7–80.0 mg, the 24-h cumulative excretion of MIBG (Table 2) was considered. The linear relationship found indicates that, under the experimental conditions studied, excretion mechanisms do not reach saturation. The clinical relevance of this is that the addition of MIBG to a therapeutic dose of ^{131}I-MIBG up to a total mass of 80 mg does not decrease the elimination rate of radioactivity from a patient's body and, therefore, does not increase that patient's radiation burden. The addition of MIBG to a therapeutic dose of ^{131}I-MIBG may increase tumour/normal tissue ratios, as has been shown in SK-N-SH xenografted mice [17].

A plot of the excretion rate versus midpoint time of

urine collection for the estimation of $t_{½,elim}$ was preferred over the procedure using the amount remaining to be excreted [14], since for the latter method complete recovery of the drug from the urine is required. In 10 of 11 patients, a high correlation was found between excretion rate and midpoint time of urine collection (r = 0.9127–0.9986). Only the plot of patient D.J. showed a weaker correlation (r = 0.7746); therefore, his relatively long $t_{½,elim}$, compared with the other children and with the amount excreted at t = 48 h (Table 2), may be less reliable.

In order to explore whether Cl_{cr} as a measure of GFR could be used as a predictor of MIBG excretion rate – and hence could be used as a tool in clinical dose adjustments – without the need for MIBG serum determinations, we investigated the possible relationship between the relative excretion rate of MIBG over the first 24 h (% dose per 24 h) and Cl_{cr}. The statistically significant correlation (r = 0.7311, P<0.02) that we found in the adult patient population was relatively weak, which does not make it a reliable predictor of MIBG excretion rate. This indicates that other processes also influence MIBG pharmacokinetics. The results of Blake *et al.* [10] suggest the importance of tubular secretion. Although many adult patients showed Cl_{cr} values (Table 2) below the normal range (70–150 ml min^{-1}), no patient showed major renal function impairment (Cl_{cr} <15 ml min^{-1}); therefore, Cl_{cr} cannot be used as a measure of GFR due to overestimation.

The Cl_{cr}/1.73 m^2 values of the children (expressed in ml min^{-1} 1.73 m^{-2}) were increased [195 (Z.L.)], or in the normal range [122 (T.R.), 102 (L.N.)]. Plotting 0–24 h MIBG excretion versus Cl_{cr}/1.73 m^2 instead of Cl_{cr} did not result in a better fit of the children's data to that of the patient population; however, it did result in the disappearance of the correlation for the adults. From the limited number of children evaluated, it can only be concluded that a possible relation exists between MIBG excretion and GFR in young patients; if such a relation does exist, it differs from that observed in adult patients.

Although the design of the present study was not appropriate for a detailed analysis of the pharmacokinetic interference of co-medications, it can be concluded from the relatively homogeneous MIBG excretion pattern in the ^{131}I-MIBG patient group, that the co-medication used by the phaeochromocytoma patients did not influence MIBG pharmacokinetics. In the MIBG patient group, patient C.O., whose 0–24 h MIBG excretion percentages were extremely low, was on an average dose of 125 μg digoxin once daily, which is known to be excreted, as MIBG, predominantly unchanged by glomerular filtration and tubular secretion. However, rather than being the result of the interaction with a standard digoxin dose, the low 0–24 h excretion of MIBG in this patient is more likely to have been the result of low creatinine clearance (Table 2).

Conclusions

After administration of therapeutic doses of ^{131}I-MIBG (3.7–7.4 GBq, 1.7–5.8 mg MIBG) to patients aged 2–73 years and suffering from various neural crest tumours, 53 ± 8.8%, 69 ± 7.8% and 83 ± 7.0% of the dose was cumulatively excreted as MIBG in the urine after 24, 48 and 72 h, respectively. The $t_{½,elim}$, as determined from urine samples taken up to 44 h after the start of infusion, ranged from 9.1 to 14.3 h (mean ± S.D.: 11.5 ± 1.7 h, n = 11). Within the dose range 1.7–80.0 mg MIBG, a significant (P<0.001) and highly correlated (r = 0.9556) linear relationship was noted between the MIBG renal excretion rate over the first 24 h (mg per 24 h) and the milligram dose of MIBG. This suggests that the addition of MIBG to a therapeutic ^{131}I-MIBG dose of up to 80 mg may not decrease the elimination rate of radioactivity from a patient's body. When the children were excluded from the analysis, a significant correlation (r = 0.7311, P<0.02) was found between the relative renal excretion rate of MIBG over the first 24 h (% dose per 24 h) and GFR, which indicates that glomerular filtration only partially determines MIBG excretion in adults and, therefore, is not a reliable predictor of MIBG excretion rate. A relation between MIBG excretion and GFR in children remains to be established.

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References

1. Hoefnagel CA, Lewington VJ. In: Murray IPC, Ell PJ, eds. Nuclear medicine in clinical diagnosis and treatment. London: Churchill Livingstone, 1994: 851–864.
2. Hoefnagel CA. Radionuclide therapy revisited. *Eur J Nucl Med* 1991; **18**: 408–431.
3. Shapiro B, Sisson JC, Wieland DM, Mangner TJ, Zempel SM, Mudgett E, Gross MD, Carey JE, Zasadny KR, Beierwaltes WH. Radiopharmaceutical therapy of malignant pheochromocytoma with [^{131}I]metaiodobenzylguanidine: Results from ten years of experience. *J Nucl Biol Med* 1991; **35**: 269–276.
4. Hoefnagel CA, de Kraker J, Voûte PA, Valdés Olmos RA. Preoperative [^{131}I]metaiodobenzylguanidine therapy of neuroblastoma at diagnosis ('MIBG *de novo*'). *J Nucl Biol Med* 1991; **35**: 248–251.
5. Hoefnagel CA, Taal BG, Valdés Olmos RA. Role of

[^{131}I]meta-iodobenzylguanidine therapy in carcinoids. *J Nucl Biol Med* 1991; **35**: 346–348.
6. Wafelman AR, Hoefnagel CA, Maes RAA, Beijnen HJ. Radioiodinated metaiodobenzylguanidine, a review of its biodistribution and pharmacokinetics, drug interactions, cytotoxicity and dosimetry. *Eur J Nucl Med* 1994; **21**: 545–559.
7. Mangner TJ, Tobes MC, Wieland DM, Sisson JC, Shapiro B. Metabolism of iodine-131 metaiodobenzylguanidine in patients with metastatic pheochromocytoma. *J Nucl Med* 1986; **27**: 37–44.
8. Wafelman AR, Hoefnagel CA, Beijnen JH. Renal excretion of (I-131)MIBG and metabolites in patients. *Pharm Weekbl Sci* 1992; **14**: F32.
9. Wafelman AR, Nortier YLM, Rosing H, Underberg WJM, Maes RAA, Beijnen JH. High-performance liquid chromatographic determination of *m*-iodobenzylguanidine in urine of cancer patients. *J Chromatogr Biomed Appl* 1993; **622**: 71–77.
10. Blake GM, Lewington VJ, Zivanovic MA, Ackery DM. Glomerular filtration rate and the kinetics of ^{123}I-metaiodobenzylguanidine. *Eur J Nucl Med* 1989; **15**: 618–623.
11. Wafelman AR, Suchi R, Hoefnagel CA, Beijnen JH. Radiochemical purity of [^{131}I]MIBG infusion fluids: A report from clinical practice. *Eur J Nucl Med* 1993; **20**: 614–616.
12. Wafelman AR, Hoefnagel CA, Beijnen JH. Chromatographic determination of the radiochemical purity of [^{131}I]MIBG infusion fluids: A comparison and discussion of the chromatographic characteristics using three different techniques. *Appl Radiat Isot* 1993; **44**: 859–867.
13. Wieland DM, Wu JL, Brown LE, Mangner TJ, Swanson DP, Beierwaltes WH. Radiolabeled adrenergic neuron-blocking agents: Adrenomedullary imaging with [^{131}I]iodobenzylguanidine. *J Nucl Med* 1980; **21**: 349–353.
14. Rowland M, Tozer TN. Clinical pharmacokinetics: Concepts and applications. Philadelphia, PA: Lea and Febiger, 1980: 88.
15. Cockroft DW, Gault MK. Prediction of creatinine clearance from serum creatinine. *Nephron* 1976; **16**: 31–41.
16. Schwartz GJ, Haycock GB, Edelmann Jr CM, Spitzer A. A simple estimate of glomerular filtration rate in children derived from body length and plasma creatinine. *Pediatrics* 1976; **58**: 259–263.
17. Rutgers M, Buitenhuis C, Voûte PA, Smets LA. High doses of unlabeled MIBG improve 'neuroblastoma/normal tissue' ratio of tracer [^{131}I]MIBG in xenografted balb/c nu/nu mice. In: Evans AE, D'Angio GJ, Knudson AG, Seeger RC, eds. Advances in neuroblastoma research 4. New York: Wiley-Liss, in press.